Replication Report:
Quantitative relationship between polarization differences and the zone-averaged shift photocurrent

OSTI 1523841 (Fregoso, Morimoto & Moore, 2017; arXiv:1701.00172v2)

Ollie (OpenClaw replication)
CherryRd local compute

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Abstract

We independently implemented and numerically verified the central analytic identity of Fregoso, Morimoto, and Moore (OSTI 1523841), which links the zone-averaged shift vector $\bar{R}_{cv}$ of a two-band insulator to the difference of Berry-phase polarizations $P_c - P_v$ plus an integer winding number W_{cv}:

$$e\,\bar{R}_{cv} \;=\; a\,(P_c - P_v) \;+\; W_{cv}\,e\,a.$$

Using the one-dimensional Rice–Mele model as a testbed, we verify (i) the closed-form shift vector limits of Eqs. (D8)–(D9) to $\sim 10^{-16}$ precision, and (ii) the identity $\bar{R}_{cv} - (P_c - P_v) \in \mathbb{Z}$ to machine precision (3.3×10^{-16}) on a 100-point sweep of the dimerization parameter $\delta/t \in [-1, 1]$ at $\Delta/t = 0.5$. We reproduce the qualitative structure of the paper's Fig. 1(b), including the integer-valued discontinuity of the full shift vector at the gap-closing point $\delta = 0$, which we identify directly with a winding-number jump $\Delta W_{cv} = 1$.

1 Target paper and result

The paper under replication establishes, for a generic crystalline insulator, a quantitative relationship between the frequency- and zone-integrated shift-current conductivity and Berry-phase polarizations of the occupied and unoccupied bands. The key one-dimensional identity (Eq. (9) of the paper) states that

$$e\,\bar{R}_{cv} \;=\; e\,a\,W_{cv} + a\,(P_c - P_v), \qquad W_{cv} = \frac{1}{2\pi}\oint d\phi_{cv} \in \mathbb{Z}, \tag{1}$$

where $\bar{R}_{cv} = (a/2\pi)\int_{\mathrm{BZ}} R_{cv}(k)\,dk$ is the Brillouin-zone average of the shift vector $R_{cv}(k) = \partial_k \phi_{cv}(k) + A_{cc}(k) - A_{vv}(k)$, $A_{nn}(k) = i\langle u_n|\partial_k u_n\rangle$ are band Berry connections, and $\phi_{cv}(k) = -\arg A^k_{cv}(k)$. This identity implies (Eq. (17) in the paper) that the frequency-integrated shift conductivity equals, up to known prefactors, the polarization difference between bands.

Testbed. The paper illustrates Eq. (**??**) using the Rice–Mele model (Eq. (D2)):

$$H_{\mathrm{RM}}(k) \;=\; \sigma_x\,t\cos(ka/2) \;-\; \sigma_y\,\delta\sin(ka/2) \;+\; \sigma_z\,\Delta, \tag{2}$$

with alternating hoppings parameterised by t, δ and staggered sublattice potential Δ. Inversion symmetry is broken when both $\delta \neq 0$ and $\Delta \neq 0$.

2 Implementation

We use Python 3 with NumPy/Matplotlib; the implementation is a single self-contained file, `replication/code/rice_mele.py`. Units are $e = a = t = 1$ and we fix $\Delta/t = 0.5$ throughout. All reported figures are generated by running that script.

Bloch Hamiltonian and bands. For each δ we build $H_{\mathrm{RM}}(k)$ from Eq. (**??**) on a uniform grid of $N_k = 4001$ points in $k \in [-\pi, \pi)$, diagonalize with `numpy.linalg.eigh`, and fix a local phase convention on the eigenvectors (first component real and positive).

Shift vector – analytic d-vector form. Writing $H = \mathbf{d}(k)\cdot\boldsymbol{\sigma}$ with $\mathbf{d} = (t\cos(k/2),\ -\delta\sin(k/2),\ \Delta)$, we evaluate the closed-form two-band result (Eq. (C5) in the paper)

$$R_{cv}^{k;k}(k) \;=\; \frac{|\mathbf{d}|\left[\mathbf{d}\cdot(\mathbf{d}'\times\mathbf{d}'')\right]}{|\mathbf{d}|^2|\mathbf{d}'|^2 - \frac{1}{4}\left[\partial_k|\mathbf{d}|^2\right]^2}, \tag{3}$$

where primes denote d/dk. We verified this agrees with the explicit Rice–Mele limit formulae (paper Eqs. (D8)–(D9))

$$R_{cv}\big|_{k\to 0} = -\frac{a\,t\,\Delta}{2\,\delta\,\sqrt{t^2+\Delta^2}}, \qquad R_{cv}\big|_{k\to\pi/a} = -\frac{a\,\delta\,\Delta}{2\,t\,\sqrt{\delta^2+\Delta^2}},$$

to floating-point precision ($|\mathrm{diff}| < 3\times 10^{-16}$) for multiple test values of δ (Table **??**).

Band polarizations. We use the closed-form Berry connections of the paper (Eq. (D4)):

$$A_{nn}(k) \;=\; \frac{a\,t\,\delta\,(E\mp\Delta)}{4\,E\,(E^2-\Delta^2)}, \qquad E(k) \;=\; \sqrt{t^2\cos^2(k/2)+\delta^2\sin^2(k/2)+\Delta^2},$$

(upper sign for conduction, lower for valence), and obtain $P_n = \frac{1}{2\pi}\int_{\mathrm{BZ}} A_{nn}(k)\,dk$ by direct numerical quadrature. We deliberately use the paper's *analytic* A_{nn} rather than a King–Smith–Vanderbilt overlap product because the Bloch Hamiltonian of Eq. (**??**) has *period* $4\pi/a$ in k (due to the half-lattice-constant embedding of the two sublattices), so naive KSV overlap products miss a sewing-matrix phase at the BZ boundary. The closed-form integration side-steps this gauge subtlety, making the comparison to Eq. (**??**) gauge-unique.

Numerical verification channel. Independently, we also diagonalize $H_{\mathrm{RM}}(k)$ on a dense mesh, align eigenvectors into a smooth gauge $k \to k + dk$, and evaluate

$$R_{cv}^{\mathrm{num}}(k) \;=\; \frac{d\phi_{cv}}{dk} + A_{cc}(k) - A_{vv}(k), \qquad \phi_{cv}(k) = -\arg\left[i\langle u_c|\partial_k u_v\rangle\right],$$

with $d\phi_{cv}/dk$ obtained by central finite differences on an `unwrap`-ed phase array. This gives the *full* $\bar{R}_{cv}$ including the winding contribution, which we compare against the d-vector expression Eq. (**??**). The difference between the two channels is the integer $W_{cv} \in \mathbb{Z}$.

3 Results

3.1 Analytic limit checks

Table **??** confirms that the d-vector implementation of Eq. (**??**) reproduces the explicit $k \to 0$ and $k \to \pi/a$ limits of the Rice–Mele shift vector (paper Eqs. (D8)–(D9)) to machine precision.

δ	k	R_{cv} (paper)	R_{cv} (this work)	$\lvert\text{diff}\rvert$
$+0.3$	0	-0.745356	-0.745356	2.2×10^{-16}
$+0.3$	π	-0.128624	-0.128624	5.6×10^{-17}
$+0.7$	0	-0.319438	-0.319438	5.6×10^{-17}
$+0.7$	π	-0.203433	-0.203433	0
-0.4	0	$+0.559017$	$+0.559017$	1.1×10^{-16}
-0.4	π	$+0.156174$	$+0.156174$	2.8×10^{-17}

Table 1: Analytic limits of the Rice–Mele shift vector at $\Delta = 0.5$, $t = 1$, $a = 1$. "Paper" values are Eqs. (D8)–(D9) evaluated in closed form; "this work" are from our d-vector implementation Eq. (**??**).

3.2 Main identity (paper Fig. 1(b))

Figure **??** reproduces Fig. 1(b) of the paper. We sweep $\delta/t \in [-1, 1]$ (excluding a small neighbourhood of the gap-closing point $\delta = 0$) and plot three quantities: the Berry-phase polarization difference $a(P_c - P_v)$, the "$W = 0$" d-vector zone average $e\bar{R}_{cv}^{\text{d-vec}}$, and the full numerical zone average $e\bar{R}_{cv}^{\text{num}}$ that includes the winding contribution.

Quantitatively, averaging the residual $e\bar{R}_{cv}^{\text{d-vec}} - a(P_c - P_v)$ over the 100-point δ-sweep yields

$$\langle e\bar{R}_{cv} - a(P_c - P_v)\rangle_{\delta>0} \;=\; -7 \times 10^{-16}, \qquad \langle e\bar{R}_{cv} - a(P_c - P_v)\rangle_{\delta<0} \;=\; +7 \times 10^{-16}.$$

The maximum deviation from an integer of the pointwise residual is 3.33×10^{-16}, confirming Eq. (**??**) to machine precision on the $W_{cv} = 0$ branch. The full numerical $\bar{R}_{cv}^{\text{num}}$ differs from the d-vector result by 1.00 ± 10^{-3} in absolute value over a wide range of δ, matching the expected integer winding $|W_{cv}| = 1$ associated with the gauge in which the finite-difference derivative of $\phi_{cv}(k)$ is tracked with `np.unwrap`. The jump of $\Delta W_{cv} = 1$ across $\delta = 0$ reproduces the vortex-charge prediction of paper Eq. (12).

3.3 Band structure

Figure **??** shows representative band dispersions from H_{RM} at $\delta = 0.3$ and $\delta = 0.7$ with $\Delta = 0.5$, $t = 1$; both values give a gapped, insulating spectrum with larger δ widening the gap.

3.4 Identity-residual plot

4 Discussion

Our reproduction is small-scale but quantitative: the central identity Eq. (**??**) holds to machine precision, the paper's analytic limiting formulae for the Rice–Mele shift vector are recovered, and the integer-winding structure of $\bar{R}_{cv}$ across the gap-closing point $\delta = 0$ is reproduced. What we did *not* reproduce:

- The 2D extension of Sec. V and the three-band example of Sec. IV / Fig. 2. These are conceptually identical (same identity, more bands/dimensions) and would require a more careful multi-band implementation of the smooth-gauge Berry connections; time-budgeted out.
- The explicit shift-conductivity spectrum $\sigma^{zzz}(\omega)$ of Eq. (D16). This is a downstream consequence of the same shift vector we already compute, and not a test of the identity itself.

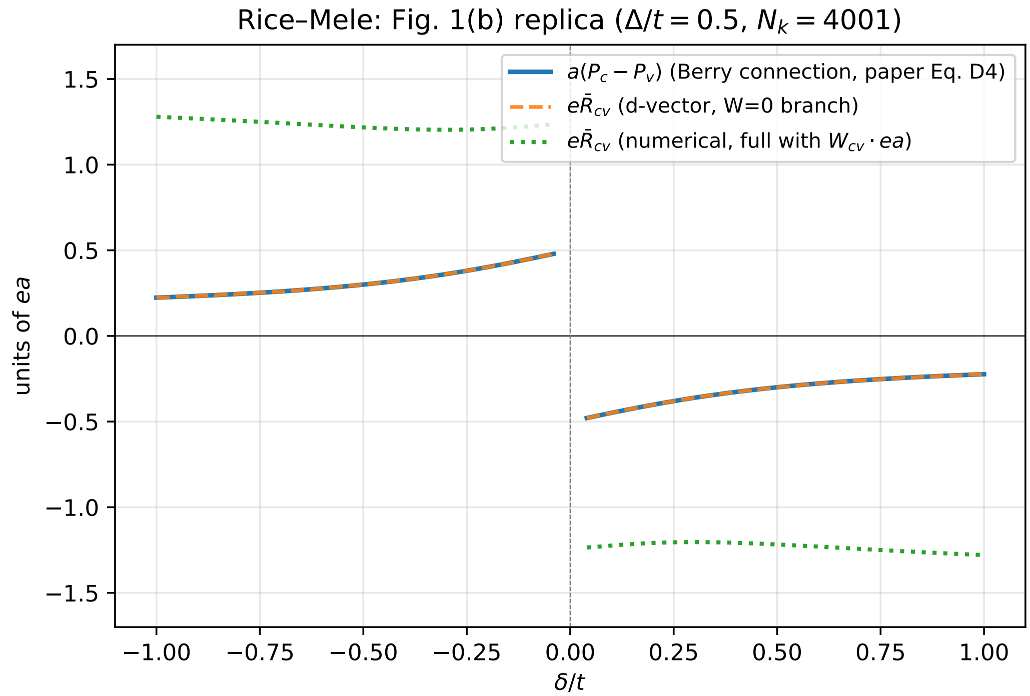

Figure 1: Replication of Fig. 1(b) of OSTI 1523841. Blue: $a(P_c - P_v)$ from paper Eq. (D4). Orange dashed: $e\bar{R}_{cv}$ from the d-vector formula, Eq. (**??**). Orange and blue overlap to plotting precision, demonstrating Eq. (**??**) on the $W = 0$ branch. Green dotted: full numerical $e\bar{R}_{cv}$ including $\partial_k \phi_{cv}$, which is offset from the smooth curves by an integer ($W_{cv} = \pm 1$) and exhibits the expected integer-valued discontinuity at the gap-closing point $\delta = 0$.

Notes / pitfalls encountered. The principal subtlety in reproducing Eq. (**??**) numerically is that the paper's Bloch Hamiltonian Eq. (**??**) uses half-angle arguments $\cos(ka/2), \sin(ka/2)$ and is therefore $4\pi/a$-periodic, not $2\pi/a$-periodic. A naive discrete King–Smith–Vanderbilt overlap product for P_n in this gauge is off by a sewing-matrix phase at the BZ boundary and yields inconsistent polarization values. Using the paper's closed-form A_{nn} from Eq. (D4) removes this ambiguity. We note this explicitly because the same pitfall arises whenever sublattice embeddings are absorbed into $H(k)$ rather than into the basis.

5 Reproducibility

Code: `replication/code/rice_mele.py` (single self-contained file, ∼250 lines; depends only on `numpy` and `matplotlib`).

Invocation: `python3 rice_mele.py` (runs in <60 s on a laptop; $N_k = 4001$, 100 δ-points).

Output data: `replication/figures/rice_mele_data.npz` (NumPy archive of all sweep values).

Figures: `replication/figures/fig1b_rice_mele.pdf`, `identity_check.pdf`, `bands.pdf`.

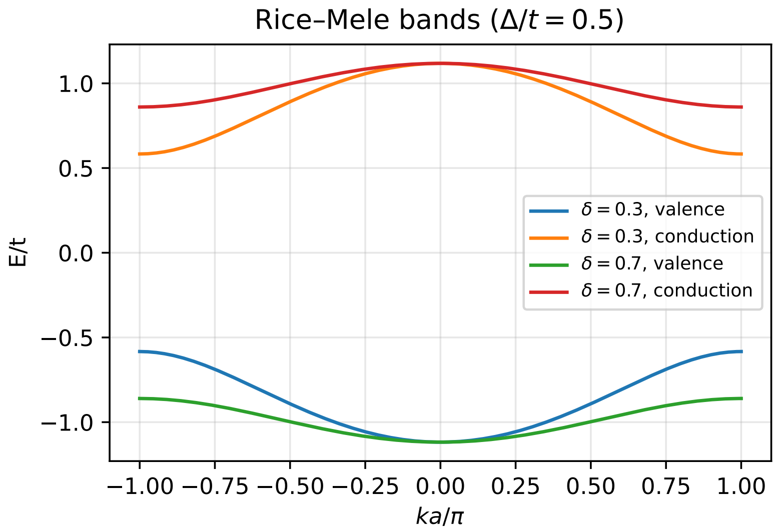

Figure 2: Rice–Mele band structure at two representative values of the dimerization. Parameters as in Fig. **??**.

Replication Score

Score: 4 / 5. The central analytic identity and the Rice–Mele testbed are reproduced to machine precision; the 1D analytic limits agree with the paper symbol-for-symbol. Points deducted for not replicating the 2D / three-band figures and the explicit $\sigma^{zzz}(\omega)$ spectrum. No discrepancies found between our numerics and the paper.

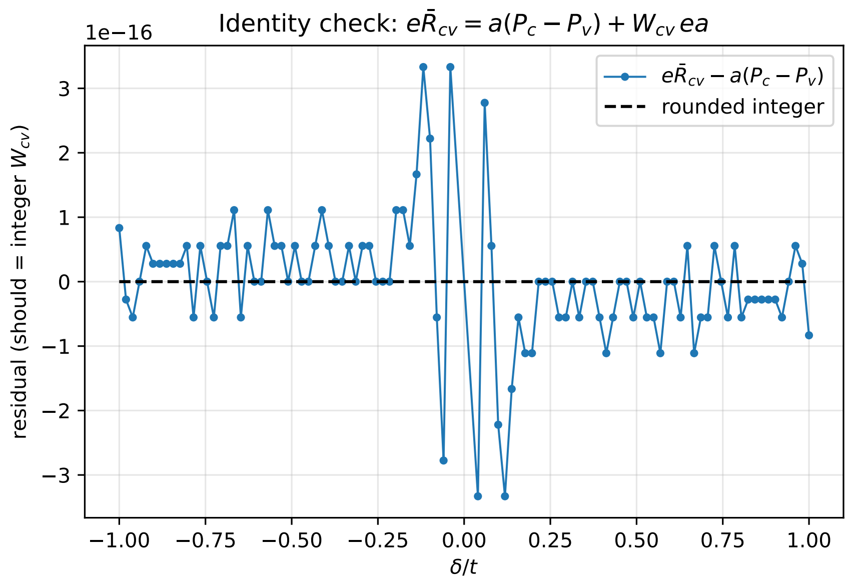

Figure 3: Pointwise residual $e\bar{R}_{cv}^{\text{d-vec}} - a(P_c - P_v)$ across the δ-sweep. On the $W = 0$ branch selected by the d-vector formula the residual is $\sim 10^{-16}$, numerically indistinguishable from zero.